

Cmod A7-35T Internal IP Peripheral Reference

Platform: Xilinx Artix-7 (xc7a35tcpg236-1) — Cmod A7-35T
Processor: MicroBlaze RISC-V
System Clock: 100 MHz (PLL from 12 MHz on-board oscillator via Clocking Wizard)
Toolchain: Vivado & Vitis 2025.2

1. Processor Core & System Infrastructure

1.1 MicroBlaze RISC-V (`microblaze_riscv_0`)

Parameter	Value
IP Version	<code>xilinx.com:ip:microblaze_riscv:1.0</code>
Instruction / Data Bus	32-bit LMB (Local Memory Bus) + AXI4 Data Port
Debug Interface	Enabled (<code>C_DEBUG_ENABLED = 1</code>)
Multiply / Divide	Enabled (<code>C_USE_MULDIV = 2</code>)
Bit Manipulation	Bitmanip A/B/C/S all enabled
Cache	16 KB I-Cache + 16 KB D-Cache, write-through, 32 B lines, cacheable range <code>0x6000_0000 – 0x6007_FFFF</code> (exact SRAM fit)

Description: The main processor core. Executes user firmware and accesses all peripherals through the AXI SmartConnect interconnect.

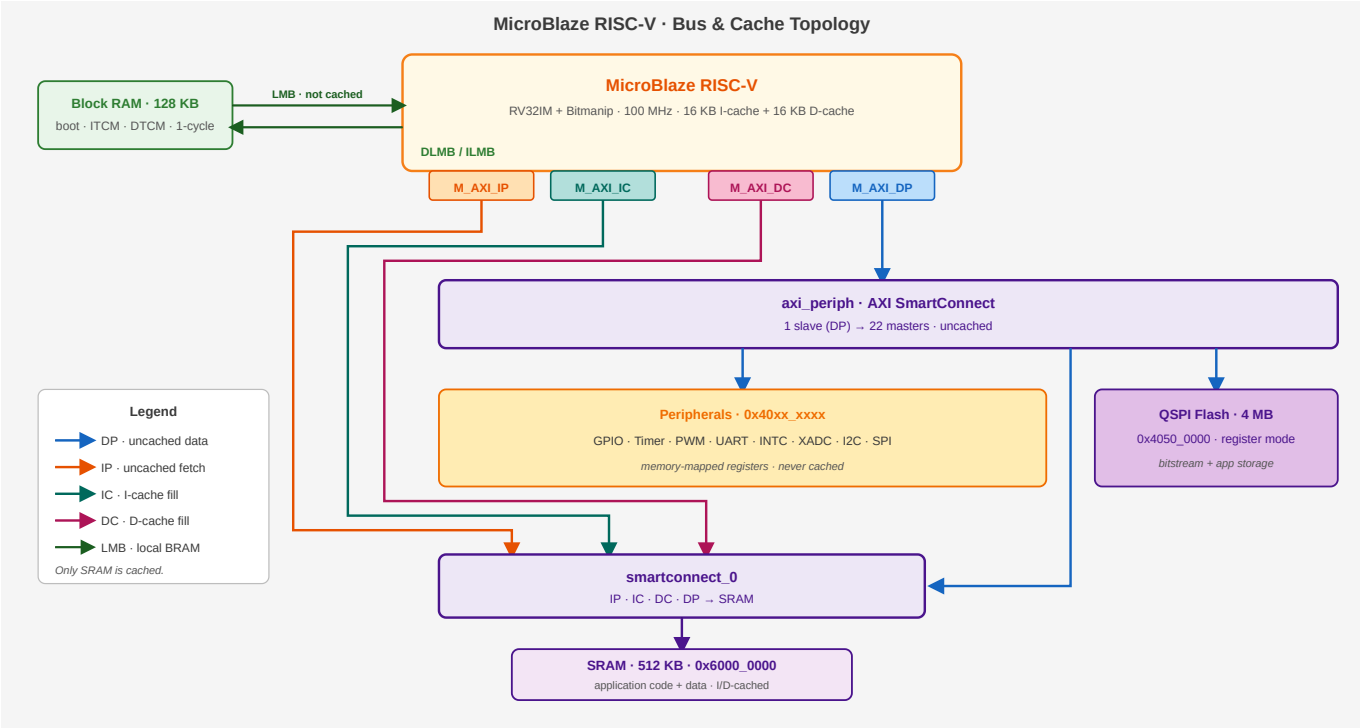
1.2 Local Memory

Parameter	Value
IP	<code>lmb_v10</code> (LMB Bus) + <code>lmb_bram_if_cntlr</code> + <code>blk_mem_gen</code>
Data Bus (DLMB)	<code>0x0000_0000 – 0x0001_FFFF</code> (128 KB)
Instruction Bus (ILMB)	<code>0x0000_0000 – 0x0001_FFFF</code> (128 KB)
Memory Type	True Dual-Port Block RAM
ECC	Disabled

Description: Instruction and data local memory implemented with FPGA Block RAM — functionally this MCU's tightly-coupled memory (TCM): the ILMB/DLMB ports play the same role as ITCM/DTCM on other MCU cores (e.g. Cortex-M7), giving 1-cycle access outside the cache path. Layout: `0x0000 – 0x7FFF` (32 KB)

holds the UART bootloader (restored from flash at every configuration); `0x8000 – 0xFFFF` (32 KB) is the application's ITCM for interrupt handlers and timing-critical code (`ITCM_FUNC` , copied out of the SRAM image by `tcm_init()` at startup); `0x10000 – 0x1FFFF` (64 KB) is the DTCM, holding the stack (top-down from `0x20000`) and `DTCM_DATA` fast data.

1.3 AXI SmartConnect (`microblaze_riscv_0_axi_periph`)



Parameter	Value
IP Version	<code>xilinx.com:ip:smartconnect:1.0</code>
Master Ports	22 (M00 – M21)
Slave Ports	1 (S00, connected to MicroBlaze M_AXI_DP)

Description: AXI interconnect crossbar that routes processor data transactions to 22 peripheral endpoints.

1.4 AXI Interrupt Controller (`microblaze_riscv_0_axi_intc`)

Parameter	Value
IP Version	<code>xilinx.com:ip:axi_intc:4.1</code>
AXI Base Address	<code>0x4040_0000</code>
Fast Interrupt	Disabled (<code>C_HAS_FAST = 0</code>)
Interrupt Sources	8 (merged via <code>ilconcat</code>)

Interrupt Mapping:

Channel	Source	Description
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Channel	Source	Description
In0	timer_0	System timer 0 interrupt
In1	timer_1	System timer 1 interrupt
In2	timer_2	System timer 2 interrupt
In3	uart_1	External UART interrupt
In4	uart_USB	USB UART interrupt
In5	INT_0_3	External GPIO interrupt (4-bit)
In6	i2c_0	I2C controller interrupt
In7	spi_0	External SPI master interrupt

1.5 Debug Module (mdm_1)

Parameter	Value
IP Version	xilinx.com:ip:mdm_riscv:1.0

Description: MicroBlaze Debug Module providing JTAG debug access for breakpoints, register inspection, and memory read/write. Its Debug_SYS_Rst signal is connected to the system reset module.

1.6 Clocking Wizard (clk_wiz_1)

Parameter	Value
IP Version	xilinx.com:ip:clk_wiz:6.0
Input Clock	12 MHz (on-board oscillator)
Output Clock	100 MHz
PLL Multiply Factor	MMCM_CLKFBOUT_MULT_F = 62.5
PLL Divide Factor	MMCM_CLKOUT0_DIVIDE_F = 7.5

Description: Uses an MMCM/PLL to multiply the 12 MHz board clock to 100 MHz for the entire system. The Locked signal indicates clock stability.

1.7 Processor System Reset (rst_clk_wiz_1_100M)

Parameter	Value
IP Version	xilinx.com:ip:proc_sys_reset:5.0
External Reset	On-board push button (Active Low)

Description: Generates synchronized reset signals (mb_reset , bus_struct_reset , peripheral_aresetn) ensuring all modules are released from reset only after the clock is stable.

2. UART Communication

2.1 USB UART (uart_USB)

Parameter	Value
IP Version	xilinx.com:ip:axi_uart16550:2.0
AXI Base Address	0x4030_0000
Address Range	64 KB (0x4030_0000 – 0x4030_FFFF)
TX Pin	J18 (via Micro-USB connector)
RX Pin	J17 (via Micro-USB connector)
Interrupt	Connected to xlconcat In4

Description: 16550-compatible UART for host PC communication through the on-board Micro-USB connector. Baud rate is software-configured via the Divisor Latch Register. At 100 MHz system clock, the divisor for 115200 baud is 54 (0x36). Typically mapped as STDIN/STDOUT.

2.2 External UART (uart_1)

Parameter	Value
IP Version	xilinx.com:ip:axi_uart16550:2.0
AXI Base Address	0x4031_0000
Address Range	64 KB (0x4031_0000 – 0x4031_FFFF)
TX Pin	J1 (DIP Pin 11)
RX Pin	K2 (DIP Pin 12)
Interrupt	Connected to xlconcat In3

Description: Second 16550 UART exposed on the DIP connector for communication with external devices (e.g., sensor modules, Bluetooth modules).

3. GPIO (General Purpose I/O)

All GPIO instances use xilinx.com:ip:axi_gpio:2.0 .

3.1 On-Board GPIO

Instance	AXI Base Address	Width	Direction	Connection	Description
board_led_2bits	0x4000_0000	2	Output	A17, C16	On-board LEDs × 2

Instance	AXI Base Address	Width	Direction	Connection	Description
<code>board_button</code>	<code>0x4001_0000</code>	1	Input	A18	On-board push button × 1
<code>board_rgb</code>	<code>0x4002_0000</code>	3	Output	B17, B16, C17	On-board RGB LED (R/G/B)

3.2 DIP Connector GPIO (4 Groups × 7-bit)

Instance	AXI Base Address	Width	DIP Pins	Description
<code>gpio_A_0_6</code>	<code>0x4003_0000</code>	7	Pin 1–7	GPIO Group A, bidirectional I/O
<code>gpio_B_0_6</code>	<code>0x4004_0000</code>	7	Pin 17–23	GPIO Group B, bidirectional I/O
<code>gpio_C_0_6</code>	<code>0x4005_0000</code>	7	Pin 42–48	GPIO Group C, bidirectional I/O
<code>gpio_D_0_6</code>	<code>0x4006_0000</code>	7	Pin 26–32	GPIO Group D, bidirectional I/O

Description: Each group is a 7-bit bidirectional port (`C_ALL_OUTPUTS = 0`). Pin direction is set through the TRI register; data is read/written via the DATA register.

3.3 External Interrupt Inputs (`INT_0_3`)

Parameter	Value
AXI Base Address	<code>0x4007_0000</code>
Width	4-bit (all inputs)
Interrupt Capability	Enabled (<code>C_INTERRUPT_PRESENT = 1</code>)
DIP Pins	Pin 8 (INTR_0), Pin 9 (INTR_1), Pin 41 (INTR_2), Pin 33 (INTR_3)

Description: Four external interrupt inputs grouped into a single AXI GPIO instance. The interrupt signal is routed to `xlconcat In5` and then to the AXI Interrupt Controller.

4. Timers & PWM

All timer instances use `xilinx.com:ip:axi_timer:2.0`.

4.1 System Timers

Instance	AXI Base Address	Mode	Interrupt	Description
<code>timer_0</code>	<code>0x4010_0000</code>	32-bit (Default)	<code>xlconcat In0</code>	General-purpose system timer

Instance	AXI Base Address	Mode	Interrupt	Description
timer_1	0x4011_0000	Default	xlconcat In1	General-purpose timer
timer_2	0x4012_0000	Default	xlconcat In2	General-purpose timer

4.2 PWM Outputs

Instance	AXI Base Address	Output Pin	DIP Pin	Description
PWM_0	0x4020_0000	J3	Pin 10	PWM Channel 0
PWM_1	0x4021_0000	W3	Pin 34	PWM Channel 1
PWM_2	0x4022_0000	W4	Pin 40	PWM Channel 2

Description: AXI Timer instances configured in PWM mode to generate square-wave outputs. Useful for LED dimming, motor speed control, buzzer tone generation, etc. Frequency and duty cycle are configured through the Timer Load Registers and the PWM enable bit.

5. External Memory

5.1 QSPI Flash (axi_quad_spi_0)

Parameter	Value
IP Version	xilinx.com:ip:axi_quad_spi:3.2
AXI_LITE Base (Control)	0x4050_0000 (64 KB)
Mode	Standard SPI controller (C_XIP_MODE = 0) — not memory-mapped
SPI Memory Type	Macronix (C_SPI_MEMORY = 4)
Interface	Quad SPI, FIFO depth 256

Description: Controls the on-board Quad-SPI NOR Flash. The controller runs in **standard (register) mode**: the full register set (control, status, TX/RX FIFO, slave select — see PG153) is exposed at 0x4050_0000 via microblaze_riscv_0_axi_periph M19, so the CPU can issue any SPI command — read (0x0B), Write Enable (0x06), Sector/Block Erase (0x20 / 0xD8), Page Program (0x02), status poll (0x05). This is what allows the UART bootloader to program application images into flash at runtime (workspace-example/boot loader/src/boot loader .c is a complete worked example, and the Vitis xSpi driver wraps the register protocol).

Flash contents are **not memory-mapped** in this mode — there is no XIP window, and code cannot execute from flash directly. The standalone-boot design instead copies the application from flash into SRAM at

power-on (see the Standalone Boot Mode guide). The 256-entry FIFO matches the flash's 256-byte page size, so one page program fits in a single transfer.

Design note: XIP mode (`C_XIP_MODE = 1` , a read-only memory-mapped window) and CPU-programmable register mode are **mutually exclusive** in this IP (PG153). This project chooses register mode: self-programming (Arduino-style `upload.py` workflow) was judged more valuable for the course than execute-in-place, and the 512 KB SRAM + I-cache serves execution instead.

5.2 SRAM / Cellular RAM (`axi_emc_0`)

Parameter	Value
IP Version	<code>xilinx.com:ip:axi_emc:3.0</code>
AXI Base Address	<code>0x6000_0000</code>
Address Range	512 KB, exact fit (0x6000_0000 – 0x6007_FFFF)
Physical Capacity	512 KB Cellular RAM

Description: External memory controller providing access to the on-board 512 KB SRAM. Suitable for large data buffers, but access latency is higher than Block RAM.

6. Analog-to-Digital Conversion (XADC)

6.1 XADC Wizard (`xadc_wiz_0`)

Parameter	Value
IP Version	<code>xilinx.com:ip:xadc_wiz:3.3</code>
AXI Base Address	<code>0x4060_0000</code>
Address Range	64 KB
Conversion Rate	500 KSPS
Sequencer Mode	Continuous

Enabled Channels:

Channel	Source	Description
VAUX4	DIP Pin 15 (G3/G2)	External analog input 0 (on-board divider: 0–3.3 V → 0–1 V)
VAUX12	DIP Pin 16 (H2/J2)	External analog input 1 (on-board divider: 0–3.3 V → 0–1 V)
Temperature	Internal	FPGA die temperature monitor
VCCINT	Internal	Core voltage monitor (1.0 V)
VCCAUX	Internal	Auxiliary voltage monitor (1.8 V)

Description: The Artix-7 built-in 12-bit ADC capable of measuring external analog signals and monitoring internal FPGA temperature and supply voltages. External input pins pass through an on-board resistive voltage divider (2.32 K Ω / 1 K Ω , ratio $\approx$ 0.301), accepting up to 3.3 V at the DIP pin.

Effective Per-Channel Sampling Rate: The sequencer continuously cycles through all 5 enabled channels (VAUX4, VAUX12, Temperature, VCCINT, VCCAUX). The aggregate conversion rate is 500 KSPS, giving each channel an effective rate of $500\text{ K} \div 5 = \mathbf{100\text{ KSPS}}$.

7. Serial Expansion Interfaces (I2C / SPI)

7.1 I2C Controller (`i2c_0`)

Parameter	Value
IP Version	<code>xilinx.com:ip:axi_iic:2.1</code>
AXI Base Address	<code>0x4070_0000</code>
SCL Frequency	100 kHz (standard mode)
SCL / SDA Pins	DIP Pin 13 (L1) / Pin 14 (L2)
Pull-ups	Weak FPGA internal pull-ups enabled in XDC; external 4.7 kΩ to 3.3 V recommended for real devices
Interrupt	Connected to <code>xlconcat In6</code>

Description: AXI IIC master for external I2C devices (sensors, EEPROMs, OLED displays). Open-drain signaling: any device may only pull the line low; the pull-up resistor returns it high. Devices are addressed by their 7-bit I2C address. Use the Vitis `XIic` driver.

7.2 External SPI Master (`spi_0`)

Parameter	Value
IP Version	<code>xilinx.com:ip:axi_quad_spi:3.2</code> (standard mode, no STARTUP)
AXI Base Address	<code>0x4080_0000</code>
SCLK	100 MHz / 16 = 6.25 MHz (fixed at synthesis, <code>C_SCK_RATIO = 16</code>)
Slave Selects	2 (<code>C_NUM_SS_BITS = 2</code>) — two devices can share the bus
Pins	DIP 35 SCLK (V3) · 36 MOSI (W5) · 37 MISO (V4) · 38 SS0 (U4) · 39 SS1 (V5)
Interrupt	Connected to <code>xlconcat In7</code>

Description: SPI master for external devices (displays, ADCs, flash modules). Full-duplex push-pull signaling — no pull-ups needed; the active device is chosen by driving its SS line low. Use the Vitis `xspi` driver.

Note: This is a *second, independent* SPI controller — do not confuse it with `axi_quad_spi_0` (`0x4050_0000`), which is dedicated to the on-board QSPI boot flash. Firmware should select controllers by instance macro (`XPAR_SPI_0_BASEADDR` vs `XPAR_AXI_QUAD_SPI_0_BASEADDR`), never by generic `XPAR_XSPI_n_*` numbering.

8. Complete Address Map

Peripheral addresses follow a class-based convention — `0x40[C]x_xxxx` , where **C** is the peripheral class (1 MB per class, 64 KB per instance). Reading an address immediately tells you what kind of device it is: class 0 = GPIO, 1 = Timer, 2 = PWM, 3 = UART, 4 = INTC, 5 = QSPI control, 6 = XADC, 7 = I2C, 8 = SPI.

AXI Base Address	Range	Peripheral	IP Type	Category
<code>0x0000_0000</code>	128K / 128K	Local Memory (BRAM)	<code>blk_mem_gen</code>	Memory
<code>0x4000_0000</code>	64K	<code>board_led_2bits</code>	<code>axi_gpio</code>	GPIO
<code>0x4001_0000</code>	64K	<code>board_button</code>	<code>axi_gpio</code>	GPIO
<code>0x4002_0000</code>	64K	<code>board_rgb</code>	<code>axi_gpio</code>	GPIO
<code>0x4003_0000</code>	64K	<code>gpio_A_0_6</code>	<code>axi_gpio</code>	GPIO
<code>0x4004_0000</code>	64K	<code>gpio_B_0_6</code>	<code>axi_gpio</code>	GPIO
<code>0x4005_0000</code>	64K	<code>gpio_C_0_6</code>	<code>axi_gpio</code>	GPIO
<code>0x4006_0000</code>	64K	<code>gpio_D_0_6</code>	<code>axi_gpio</code>	GPIO
<code>0x4007_0000</code>	64K	<code>INT_0_3</code>	<code>axi_gpio</code>	Interrupt
<code>0x4010_0000</code>	64K	<code>timer_0</code>	<code>axi_timer</code>	Timer
<code>0x4011_0000</code>	64K	<code>timer_1</code>	<code>axi_timer</code>	Timer
<code>0x4012_0000</code>	64K	<code>timer_2</code>	<code>axi_timer</code>	Timer
<code>0x4020_0000</code>	64K	<code>PWM_0</code>	<code>axi_timer</code>	PWM
<code>0x4021_0000</code>	64K	<code>PWM_1</code>	<code>axi_timer</code>	PWM
<code>0x4022_0000</code>	64K	<code>PWM_2</code>	<code>axi_timer</code>	PWM
<code>0x4030_0000</code>	64K	<code>uart_USB</code>	<code>axi_uart16550</code>	Communication
<code>0x4031_0000</code>	64K	<code>uart_1</code>	<code>axi_uart16550</code>	Communication
<code>0x4040_0000</code>	64K	<code>axi_intc</code>	<code>axi_intc</code>	System
<code>0x4050_0000</code>	64K	<code>axi_quad_spi_0</code> (<code>AXI_LITE</code> , full register set)	<code>axi_quad_spi</code>	Memory
<code>0x4060_0000</code>	64K	<code>xadc_wiz_0</code>	<code>xadc_wiz</code>	ADC

AXI Base Address	Range	Peripheral	IP Type	Category
0x4070_0000	64K	i2c_0 (DIP 13/14)	axi_iic	Communication
0x4080_0000	64K	spi_0 (external master, DIP 35–39)	axi_quad_spi	Communication
0x6000_0000	512K	axi_emc_0 (exact physical fit, I/D-cached)	axi_emc	Memory
